Vremenski kontinuirana Fourierova TRANSFORMACIJA

(eng „CTFT)

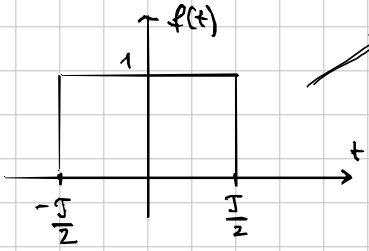
CTFT $F(j\omega)$

$$F(j\omega)=\int_{-\infty}^{\infty} f(t)\cdot e^{-j\omega t}\,dt \quad \text{CTFT}$$

$$E=\int_{-\infty}^{\infty} |f(t)|^2\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(j\omega)|^2\,d\omega$$

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty} F(j\omega)e^{j\omega t}\,dt \quad \text{ICTFT}$$

Primjer:



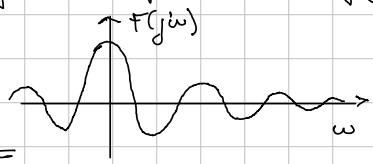
$$F(j\omega)=\int_{-\infty}^{\infty} \overbrace{1}^{f(t)}\cdot e^{-j\omega t}\,dt = \frac{1}{-j\omega}e^{-j\omega t}\Big|_{-\frac{T}{2}}^{\frac{T}{2}}$$

$$=\frac{1}{-j\omega}\left(e^{-j\omega\frac{T}{2}}-e^{+j\omega\frac{T}{2}}\right)$$

$$=\frac{1}{-j\omega}(-)2j\sin\left(\omega\frac{T}{2}\right)=\frac{+2}{\omega}\sin\left(\omega\frac{T}{2}\right)$$

$$F(j\omega)=T\cdot\frac{\sin\left(\omega\frac{T}{2}\right)}{\omega\frac{T}{2}}$$ → sinc funkcija: sinus kojem se amplituda smanjuje

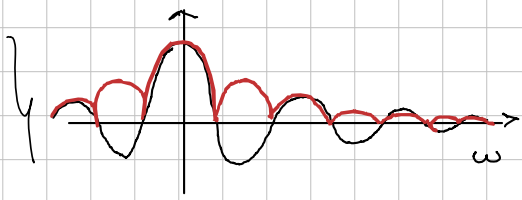
$$F(j\cdot 0)=\int_{-\infty}^{\infty} 1\cdot e^{-j\cdot 0\cdot t}\,dt=\int_{-T/2}^{T/2} 1\,dt = t\Big|_{-\frac{T}{2}}^{\frac{T}{2}} = \underline{\underline{T}}$$



=> amplitudni spektar

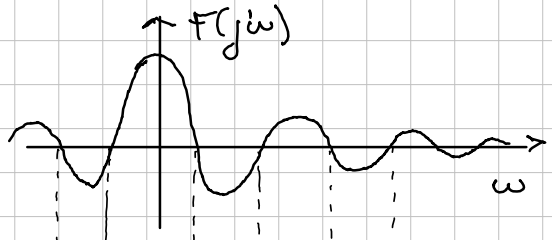
poz ostaje poz

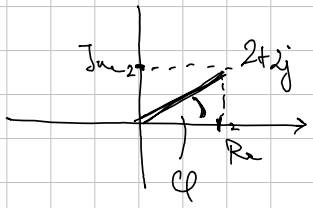
neg se zrcali gore

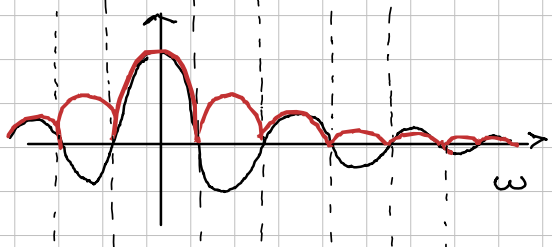


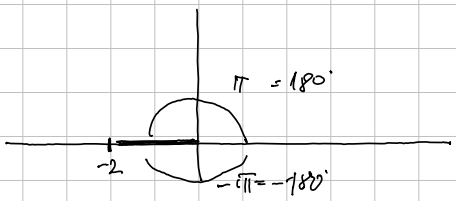
⇒ fazno fr. spektar

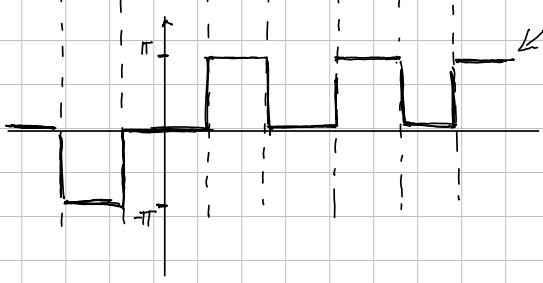
→ ako imamo samo ampl. sp. ne možemo u potpunosti jer treba fazu prikazati to zrcaljenje



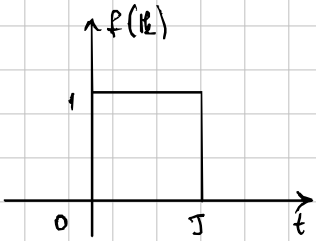








Zadatak:



$$F(j\omega) = \int_0^T 1 \cdot e^{-j\omega t}\, dt = \frac{e^{-j\omega t}}{-j\omega}\Big|_0^T$$

$$= \frac{1}{j\omega}\left(e^{-j\omega T} - 1\right) = \frac{1}{j\omega} \cdot e^{-j\omega \frac{T}{2}} \left(e^{-j\omega \frac{T}{2}} - e^{j\omega \frac{T}{2}}\right)$$

$$= \frac{1}{\cancel{j}\omega} e^{-j\omega \frac{T}{2}} \cdot (\cancel{-}2\cancel{j}) \sin\left(\omega \frac{T}{2}\right) = \underbrace{T e^{-j\omega \frac{T}{2}} \frac{\sin\left(\omega \frac{T}{2}\right)}{\omega \frac{T}{2}}}$$

$F(j\omega)$ za $\omega = 0$ $\rightarrow$ $F(j0) = \int_0^T 1\, dt = \underline{\underline{T}}$

- u prošlom zadatku smo dobili isto T za 0

! ali za ostale omega prije smo dobili sinc, odnosno $T \frac{\sin \omega \frac{T}{2}}{\omega \frac{T}{2}}$

a sada nam se pojavio član $e^{-j\omega \frac{T}{3}}$ čija je aps vrijednost 1

pa je amplitudna karakteristika <u>NEPROMJENJENA!</u>

=> ovaj pravokutnik je pomaknut u vremenu

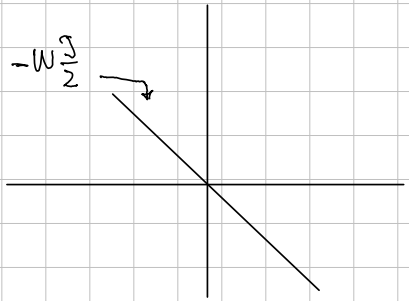
↳ mijenja se fazna karakteristika

Faza:

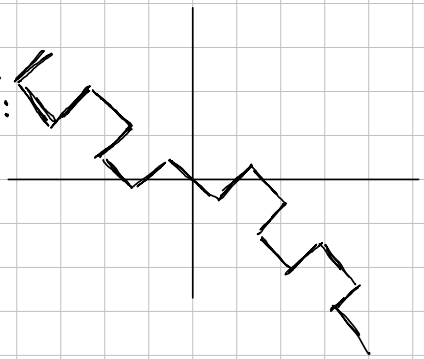
$$\measuredangle F(j\omega) = \measuredangle T \frac{\sin\left(\omega \frac{T}{2}\right)}{\omega \frac{T}{2}} - \omega \frac{T}{2}$$

bez j!

→ minus jer minus je tamo, ali općenito jer pišemo brojnik − nazivnik



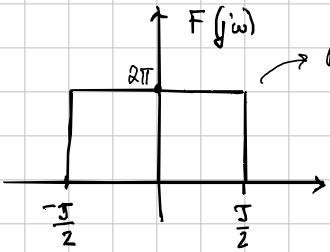
=> sveukupna fazna:



Zadatak:

CTFT $\quad F(j\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$

ICTFT $\quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) \cdot e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} 2\pi e^{j\omega t} d\omega$

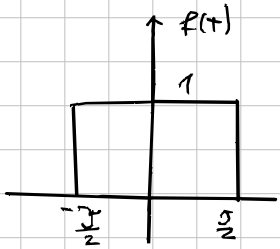


pravokutnik u frekv. domeni

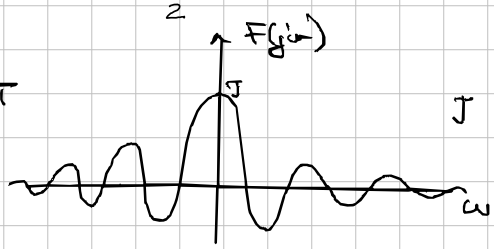
$$= \frac{1}{\cancel{2\pi}} \cdot \frac{\cancel{2\pi}}{jt} e^{j\omega t} \Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}} = \frac{e^{j\frac{\tau}{2}t} - e^{-j\frac{\tau}{2}t}}{jt}$$

$$f(t) = \frac{1}{jt} \cdot (-2) j \sin \frac{\tau}{2} t = \frac{2}{t} \sin \frac{\tau}{2} t$$

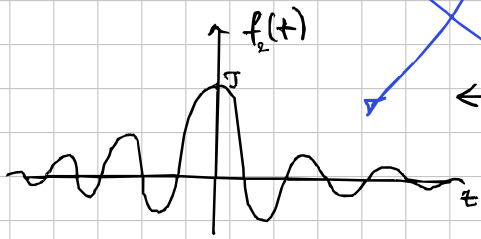
$$f(t) = \tau \frac{\sin \frac{\tau}{2} t}{\frac{\tau t}{2}} \rightarrow \text{sinc}$$



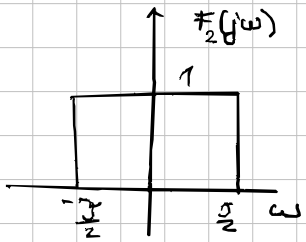
CTFT



$$\tau \frac{\sin \frac{\tau}{2} t}{\frac{\tau}{2} t}$$



ICTFT



SVOJSTVO DUALNOSTI

$$f(t) \longrightarrow F(j\omega)$$

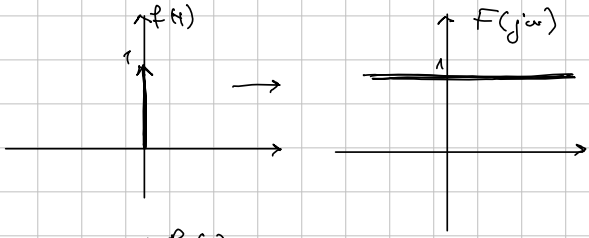
$$f_2(t) = F(jt) \longrightarrow 2\pi f(-\omega)$$

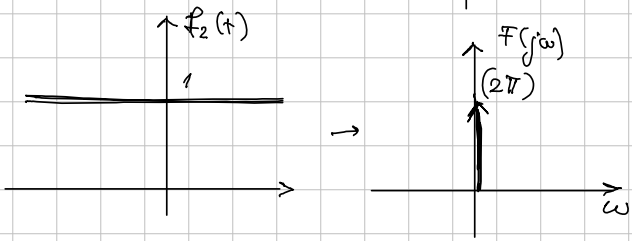
Zadatak: * ako ne vrijede Dirichletovi uvjeti → ne možemo primjeniti transformaciju

Fourierova transformacija od impulsa:

$f(t) = \delta(t)$

$$F(j\omega) = \int_{-\infty}^{\infty} \underbrace{\delta(t)}_{\text{samo za } t=0 \to 1} \cdot \overset{=1}{e^{-j\omega t}} dt = 1$$





$f(t) \longrightarrow F(j\omega)$

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